GR CARBONS

Calculation formulas

Scores on the plant dashboard · 15 Oct 2026

These are the formulas the screens use. A percent is shown to one decimal place. When the denominator is zero, or a measurement has not finished, the score stays blank and the screen says Not Evaluated. A missing measurement is never treated as zero. These scores do not create or change a bag.

Plant yield

Plant yield % = finished kilograms ÷ raw material charged to production × 100.

Finished kilograms are the stocked lot quantity plus the last dispatched kilograms on a dispatched lot. When that total is still zero, the screen uses finished-goods stock plus every dispatched kilogram. Charged kilograms are the raw material released to production. The dial arc stops at 100. The number itself can show above 100. While material is only in process, and nothing is stocked or dispatched, yield stays blank.

Material in plant

Material in plant = charcoal + awaiting kiln + in process + in stock. Dispatched kilograms stay outside this total.

Part	What is added
Charcoal	Received kilograms that are not released yet.
Awaiting kiln	On a released lot that has not started: lot kilograms minus dispatched kilograms minus kilograms already In Stock.
In process	The same remainder on a lot that is on the kiln or the shop floor.
In stock	Grade stock. When that is empty, stocked lots, then finished-goods stock.

Supplier efficiency

The overview dial is the plant total. The Raw Material tab ranks each supplier a login can see. A missing part is left out and the remaining weights are scaled to 1. The figures use all receipts to date.

Part	Formula
Usable recovery	100 × kilograms released to production ÷ plant weighbridge kilograms.
Weigh match	100 × plant kilograms ÷ supplier DC kilograms.
Weigh score	100 minus the gap between weigh match and 100. The score cannot go below 0.
Lab pass	100 × lab approvals ÷ lab decisions.
Efficiency	0.5 × recovery + 0.3 × lab pass + 0.2 × weigh score. A missing part is left out and the remaining weights are scaled to 1.

Overall process efficiency

Process input = our weight, else the inward quantity, added up for inward orders that already have a graded outward bag. Each loss % = 100 × loss kilograms ÷ process input. Actual efficiency % = 100 minus the sum of those loss percentages. The dial uses the streams that have finished. A stream still in process stays off the number. There is no approved loss standard, so the reference stays Not Evaluated. This score is not plant yield.

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Read-only scores. They do not change a bag.

Stream	Loss kilograms
Stones	Outward bag weight graded Stones. The rotary name is ignored. No Stones bag means Not Evaluated.
Unburnt	Outward bag weight graded Unburnt. No Unburnt bag means Not Evaluated. Both stay off the kiln.
Kiln	For each kiln, feed is due when the minutes since load reach that kiln’s residence time. Loss = due kilograms minus that kiln’s output. Feed still inside residence stays in process. A kiln with feed and no residence makes the kiln Not Evaluated. Output is not shared across kilns.
Screening, de-dusting, crushing, de-magnetisation, blending	Loaded kilograms minus the output bags that name those loaded bags. A group still on the loader stays in process until every bag has passed its Masters minutes. De-dusting, de-magnetisation, and blending finish when the output bag is saved. An unknown input weight makes that machine Not Evaluated.

Destoning and material handling are stored in Masters and are not loss lines. The minutes below are the top of the 500 kg estimate, so a bag still inside the estimate stays in process. Crushing and blending have no minutes until the plant sets them.

Machine	Minutes from load to output
Screening	20
De-dusting	10
Destoning	15 (stored only)
De-magnetisation	5
Material handling	10 (stored only)
Crushing	Blank
Blending	Blank

Process tracking

Score	Formula
Mesh compliance	100 × screening tags whose grade is on that mesh in Masters ÷ screening tags that have a mesh and a grade. An empty mesh list stays Not Evaluated. A refused mesh is not stored.
Destoner CTC	100 × saved destoner loads ÷ saved destoner loads. A saved load already matched the destoner CTC. No load stays Not Evaluated. CTC is rounded to the Masters decimals, then the two values must be equal.
Deviation	100 × runs with a deviation ÷ evaluated runs. A deviation is a screening grade off its mesh, or a kiln output whose measured CTC does not match the work order. Evaluated runs are those screening tags, destoner loads, and saved kiln-output quality rows. Stones marked Yes is not a deviation.

Machines

Score	Formula
Kiln designed kg for the month	Masters monthly capacity × (60 ÷ residence minutes), rounded, and at least 1 kg. Masters capacity is the output at 60 minutes. No residence uses the Masters capacity. Other machines use their Masters monthly capacity. The fallback is 15,000 kg.
Utilization	100 × output saved this calendar month ÷ designed kilograms for the month. Green is 90% and above. Amber is 70% up to green. Red is below 70%. Masters can change these bands.
Kiln cycle	Elapsed hours ÷ (residence minutes ÷ 60), capped at 100, while that kiln is loaded and a residence time is set.
Kilns running	100 × kilns marked running ÷ the number of kilns.

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Month-to-date output uses the calendar month on the screen. Shop-floor lists for today use the plant day in Asia/Kolkata. The dashboard keeps history so a past bag still counts in the month.

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